THE MALAWI NATIONAL EXAMINATIONS BOARD

2009 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

MATHEMATICS

Subject Number: M131/II

Time Allowed: 2 h 30 mins

Monday, 19 October **8.30 - 11.00 am**

PAPER II

(100 marks)

Instructions:

1. **This paper contains 6 pages. Please check.**

2. Answer **all** the **six** questions in Section **A** and any **three** questions from Section **B**.

3. The maximum number of marks for each answer is indicated against each question.

4. Mathematical tables and answer books are provided.

5. Calculators may be used.

6. Used graph paper and/or supplementary sheets must be tied together inside the answer book with a string.

7. **All working must be clearly shown**; it should be done on the same sheet as the rest of the answers.

8. Write your **Examination Number** on top of each page of your Answer Book.

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Section A (55 marks)

Answer **all** the **six** questions in this section.

1. a. Given that $\log_{10} 5 = 0.6990$, without using a calculator or tables, evaluate $\log_{10} 2$. **(3 marks)**

b. The last term in an arithmetic progression (AP) 2, 5, 8, ... is 296. Calculate the number of terms. **(4 marks)**

2. a. Given that $x = 9$, $y = 4$ and $z = 3$, evaluate $\sqrt{\dfrac{5y^2 + y^0}{xz^4}}$. **(4 marks)**

$\dfrac{5(4)^2 + 4^0}{(9 \times 3)^4}$

b. Given that $A = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 2 & -5 \end{pmatrix}$, find $2A - B^2$. **(5 marks)**

$80 + 1$

$\begin{pmatrix} 0 & 1 \\ 2 & -5 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & -5 \end{pmatrix}$

$\begin{pmatrix} 6 & -2 \\ -2 & 23 \end{pmatrix}$

$\dfrac{81}{531.6}$

$\dfrac{9}{23.1}$

$= 0.4$

3. a. Simplify $\dfrac{\sqrt{n^3} - \sqrt{9n}}{\sqrt{n}}$ in its simplest form. **(4 marks)**

b. **Figure 1** shows a speed-time graph of an object.

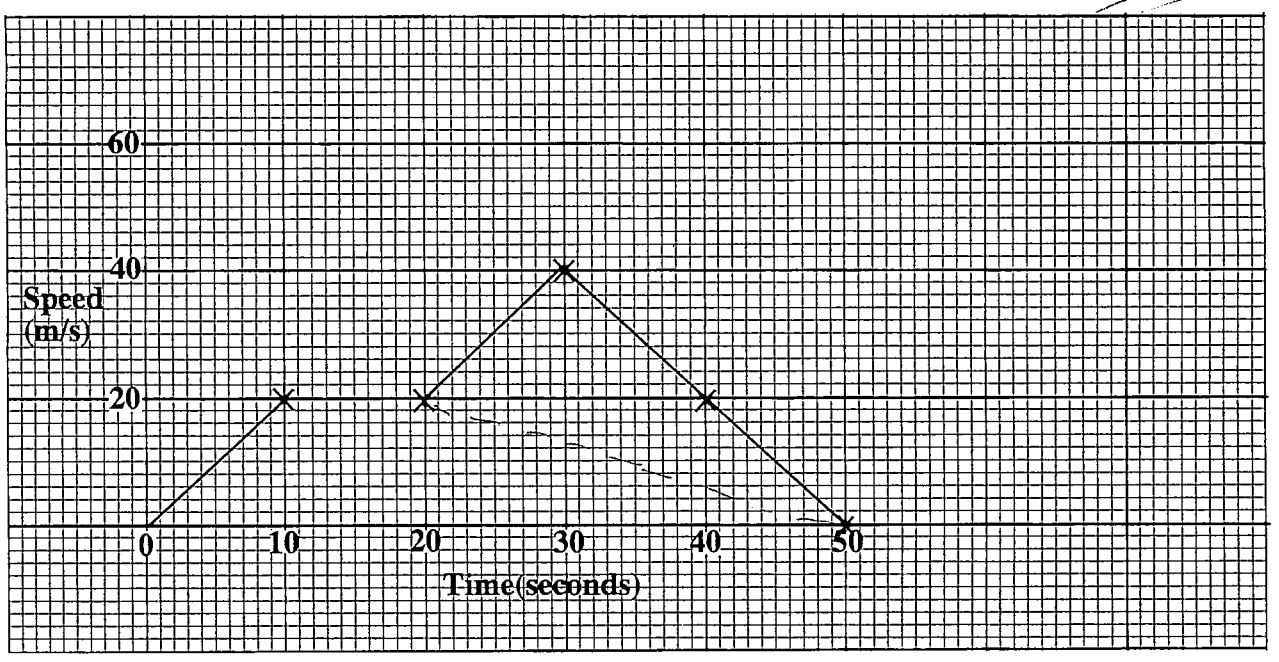


Figure 1

Calculate the distance covered by the object. **(5 marks)**

Continued/...

$2y^3 \times x\,2y^3 = \frac{5-2y^3}{5-2y^3}$

4. a. Given that $xy^3 = 5 - 2y^3$, make y the subject of the formula. **(4 marks)**

b. **Figure 2** shows a circle **MPQ** centre **O**. **TP** is a tangent to the circle at **P** and **TMRO** is a straight line.

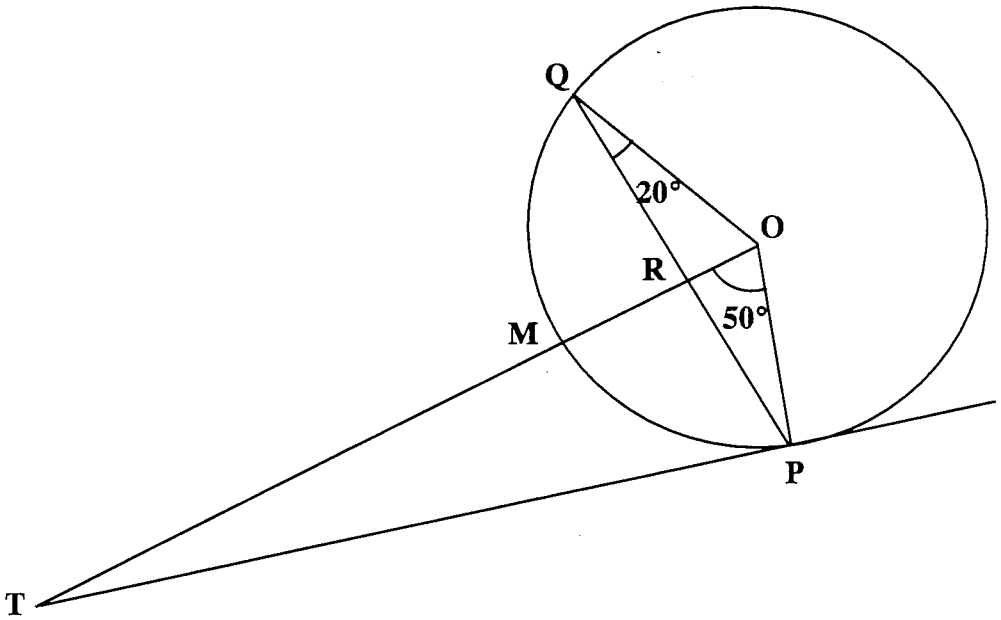


Figure 2

If angle **RQO** = 20° and angle **TOP** = 50°, show that **TP = TR.** **(6 marks)**

5. a. Simplify $\frac{t}{t^2 - at}$ in its simplest form. **(3 marks)**

$\frac{t}{t^2-at} = \frac{t}{t(t-a)}$

b. The ages of five students at Mayeso Secondary School are 14, 16, 19, 20 and 21. Calculate the variance, showing all your working. **(7 marks)**

6. a. Given that the roots of a quadratic equation in y are -5 or 3. Formulate the equation in the form $ay^2 + by + c = 0$. **(4 marks)**

$25a + 5b + c = 0$

$a(5)^2 + b(5) + c = 0$

$25a +$

b. Using a ruler and pair of compasses only, construct in the same diagram:

(i) a triangle ABC in which AB = 10 cm, AC = 7 cm and angle BAC = 60°.

(ii) two points M and N which lie on the circumference of circle ABC which are equidistant from B and C.

(iii) measure and state the length of MN. **(6 marks)**

Continued/...

Section B (45 marks)

Answer any **three** questions from this section.

7. a. **Figure 3** shows a circle **ACB** Centre **O**. **OM** = $3x$ cm, **MC** = $2x$ cm and angle **OMB** = 90°

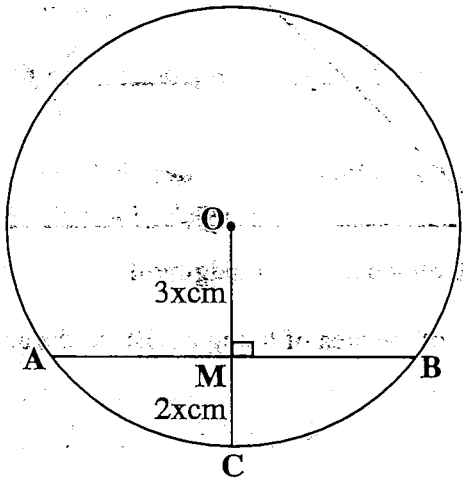


Figure 3

Find **AB** in terms of x. **(5 marks)**

b. Solve the simultaneous equations:

$$x-3y-1=0$$
$$x^2-8y^2-y+5=0.$$

(10 marks)

8. a. Given that $f(x)=x^2+1$ and $g(x)=kx+n$ where k and n are constants; and $f(o)=g(o)$ and $g(2)=15$, calculate the values of k and n. **(6 marks)**

b. A company hires two types of minibuses A and B to carry its employees. Type A carries 16 passengers and type B carries 8 passengers. The company requires atmost 10 minibuses to carry a maximum of 128 employees every day.

(i) If x represents number of the 16-passenger minibuses and y represents number of the 8-passenger minibuses, write down **two** inequalities in addition to $x \geq 0$ and $y \geq 0$.

(ii) Using a scale of 2 cm to represent 2 units on both axes, draw graphs to show the region represented by the inequalities.

(iii) Use your graph to find the number of minibuses of each type the company must hire to meet its target of transporting 128 employees at once using a minimum number of type A minibuses. **(9 marks)**

Continued/...

9. a. **Figure 4** shows triangle **ADE** in which **AC** = 5 cm, **CE** = 3 cm and **BC** is parallel to **DE**. **B** is on **AD** and **C** is on **AE**.

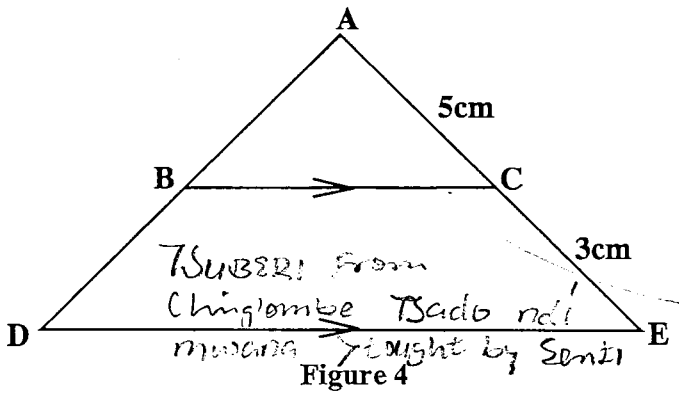


Figure 4

Calculate the ratio of the area of triangle **ABC** to the area of trapezium **BCED.** **(7 marks)**

b. Given that $x+2$ is a factor of $2x^3-3x^2-11x+6$, solve the equation $2x^3-3x^2-11x+6=0$. **(8 marks)**

10. a. Point A lies on the x-axis at -4 and point B lies on y-axis at 2. Find the equation of the straight line through A and B in the form $y = mx + c$. **(5 marks)**

b. **Figure 5,** shows a prism **ABCDEF**. **AB** = 30 m, **BC** = 10 m, **CF** = 5 m and angle **BCF** = 90°.

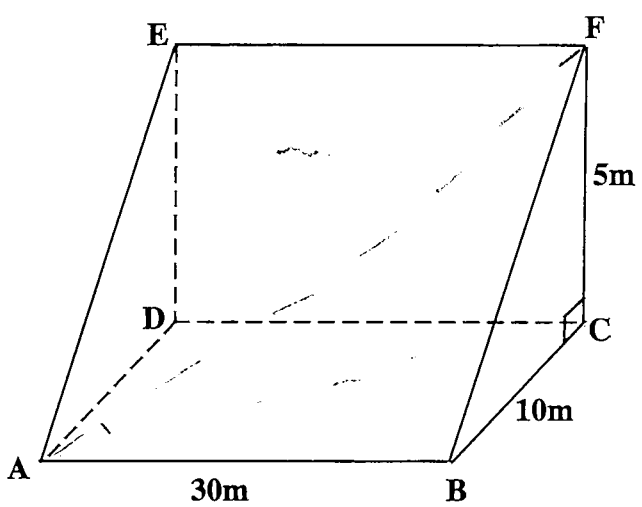


Figure 5

Calculate:

(i) the volume of the prism. $30 \times 10 \times 5 = 1500m^3$

(ii) angle **FAC** to the nearest degree. **(10 marks)**

Continued/...

11. a. Solve the equation $3x^2 - 7x - 2 = 0$, giving your answer correct to 2 decimal places. **(7 marks)**

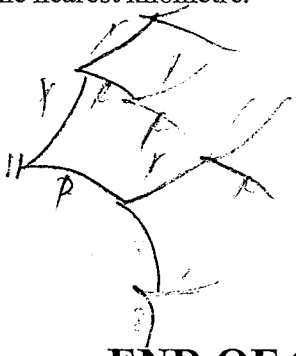
b. A variable y is partly constant and partly varies as x. When $x = 5$, $y = 4$ and when $x = 9$, $y = 8$. Find y when $x = 1$. **(8 marks)**

12. a. A set of three cards numbered 1, 2 and 3 respectively, are placed in a bag. Draw a tree diagram and use it to find:

(i) how many two-digit numbers can be formed by selecting with replacement two cards with different digits in any order.

(ii) the probability that any two-digit number formed with different digits is even. **(7 marks)**

b. A lorry leaves point A to point B travelling on a bearing of 035° for 50 km. It then turns due East travelling at 25 km/h for 4 hours and reaches point C. Calculate how far the lorry is from the starting point, leaving your answer correct to the nearest kilometre. **(8 marks)**



END OF QUESTION PAPER

NB: This paper contains 6 pages.